Activity 3: CMOS Manufacturing Flow and Control Gates

Course: Semiconductor Device and Process Integration Masterclass

Course code: C605

Duration: 50 minutes

PPT alignment: Trainer Slides v1.0, concept slides 242-280 and Activity 3 overview slide 418.

Workplace scenario

A product transfer team must turn a high-level CMOS description into a controlled manufacturing flow for a pilot lot.

Goal

Outline a coherent CMOS process flow and show where device physics constrains sequencing.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Start from incoming silicon and place isolation, well formation, gate stack, source-drain, contacts and interconnect in order.
2. Insert the lithography, etch, clean, deposition, implantation, anneal and CMP modules required by each structural change.
3. Add hold points for overlay, critical dimension, film thickness, sheet resistance and defect inspection.
4. Identify three sequence dependencies that cannot be reversed and explain the physical reason.
5. Trace one wafer through the completed flow and check that every layer or junction has a formation and verification step.

Required evidence

One-page CMOS process flow with control gates and three device-physics sequence rationales.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] Flow begins with incoming wafer and ends with test-ready die

[ ] Five required control gates are present

[ ] Three irreversible dependencies are explained

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.